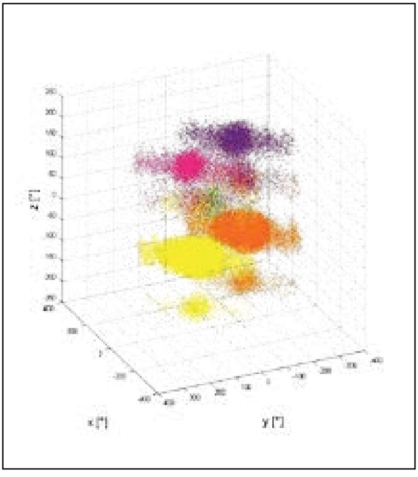
becomes unsuitable for processing big data which sometimes entails huge samples that could even include a population in its entirety.

To tackle this issue, new statistical procedures are designed. Many of these procedures have either unknown run times or long run times that make the procedure impractical if the data is seriously huge. When faced with such situations, scientists may have to resort to ad hoc methodologies many of which could lead to drawing the wrong conclusions.

2. The issue of "high dimensionality"

High dimensionality refers to the property by which big data typically contains a host of multiple information about individual parameters that are sampled. And the more "dimensions" that the samples have the chances of finding spurious links between different data-correlations that are nothing but flukes are also higher.

For instance, a medical study could find a link between the success of a medicine with the patient's height. The reality might be that when the big data includes information on parameters including height, body weight, eye colour, shoe size and more some connections may appear to be important just by chance.



Big Data can handle multidimensional data

3. The challenge of overcoming biases

What comes to be known as big data, in fact is often garnered by combining information from multiple sources-something that happens at various times and by using different technologies or methods. This heterogeneity forces the data scientists to design more adaptive procedures to analyse the data. In other words, novel statistical thinking and methods for computation are called for. This in turn demands new ways of looking at the very idea of how science should work or even what science is-a paradigm shift in mindset which is perhaps the biggest challenge of all. After all, resistance to change isn't something the